```
!pip install pandas numpy matplotlib seaborn
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/Retail and wherehouse Sale.csv')
```

```
df.head(10)
```

	YEAR	MONTH	SUPPLIER	ITEM CODE	ITEM DESCRIPTION	ITEM TYPE	RETAIL SALES	RETAIL TRANSFERS	WAREHOUSE SALES
0	2020	1	REPUBLIC NATIONAL DISTRIBUTING CO	100009	BOOTLEG RED - 750ML	WINE	0.00	0.0	2.0
1	2020	1	PWSWN INC	100024	MOMENT DE PLAISIR - 750ML	WINE	0.00	1.0	4.0
2	2020	1	RELIABLE CHURCHILL LLLP	1001	S SMITH ORGANIC PEAR CIDER - 18.7OZ	BEER	0.00	0.0	1.0
3	2020	1	LANTERNA DISTRIBUTORS INC	100145	SCHLINK HAUS KABINETT - 750ML	WINE	0.00	0.0	1.0
4	2020	1	DIONYSOS IMPORTS INC	100293	SANTORINI GAVALA WHITE - 750ML	WINE	0.82	0.0	0.0
5	2020	1	KYSELA PERE ET FILS LTD	100641	CORTENOVA VENETO P/GRIG - 750ML	WINE	2.76	0.0	6.0
6	2020	1	SANTA MARGHERITA USA INC	100749	SANTA MARGHERITA P/GRIG ALTO - 375ML	WINE	0.08	1.0	1.0
7	2020	1	BROWN-FORMAN BEVERAGES WORLDWIDE	1008	JACK DANIELS COUNTRY COCKTAIL SOUTHERN PEACH -...	BEER	0.00	0.0	2.0
					KNOB CREEK BOURBON 9YR -				

```
df.shape
```

```
(30000, 9)
```

```
df.columns
```

```
Index(['YEAR', 'MONTH', 'SUPPLIER', 'ITEM CODE', 'ITEM DESCRIPTION',
       'ITEM TYPE', 'RETAIL SALES', 'RETAIL TRANSFERS', 'WAREHOUSE SALES'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   YEAR              30000 non-null  int64
 1   MONTH             30000 non-null  int64
 2   SUPPLIER          29967 non-null  object
 3   ITEM CODE         30000 non-null  object
 4   ITEM DESCRIPTION  30000 non-null  object
```

```
 5   ITEM TYPE         30000 non-null  object
 6   RETAIL SALES      29999 non-null  float64
 7   RETAIL TRANSFERS  30000 non-null  float64
 8   WAREHOUSE SALES   30000 non-null  float64
dtypes: float64(3), int64(2), object(4)
memory usage: 2.1+ MB
```

```
df.isnull().sum()
```

	0
YEAR	0
MONTH	0
SUPPLIER	33
ITEM CODE	0
ITEM DESCRIPTION	0
ITEM TYPE	0
RETAIL SALES	1
RETAIL TRANSFERS	0
WAREHOUSE SALES	0

dtype: int64

```
df = df.dropna()
```

```
df['MONTH_NAME'] = pd.to_datetime(df['MONTH'], format='%m').dt.month_name()
```

```
df['DATE'] = pd.to_datetime(df['YEAR'].astype(str) + '-' + df['MONTH'].astype(str))
```

```
df['TOTAL SALES'] = df['RETAIL SALES'] + df['WAREHOUSE SALES']
```

```
df.head(10)
```

	YEAR	MONTH	SUPPLIER	ITEM CODE	ITEM DESCRIPTION	ITEM TYPE	RETAIL SALES	RETAIL TRANSFERS	WAREHOUSE SALES	MONTH_NAME	DATE	TOTAL SALES
0	2020	1	REPUBLIC NATIONAL DISTRIBUTING CO	100009	BOOTLEG RED - 750ML	WINE	0.00	0.0	2.0	January	2020-01-01	2.00
1	2020	1	PWSWN INC	100024	MOMENT DE PLAISIR - 750ML	WINE	0.00	1.0	4.0	January	2020-01-01	4.00
2	2020	1	RELIABLE CHURCHILL LLLP	1001	S SMITH ORGANIC PEAR CIDER - 18.7OZ	BEER	0.00	0.0	1.0	January	2020-01-01	1.00
3	2020	1	LANTERNA DISTRIBUTORS INC	100145	SCHLINK HAUS KABINETT - 750ML	WINE	0.00	0.0	1.0	January	2020-01-01	1.00
4	2020	1	DIONYSOS IMPORTS INC	100293	SANTORINI GAVALA WHITE - 750ML	WINE	0.82	0.0	0.0	January	2020-01-01	0.82
5	2020	1	KYSELA PERE ET FILS LTD	100641	CORTENOVA VENETO P/GRIG - 750ML	WINE	2.76	0.0	6.0	January	2020-01-01	8.76
6	2020	1	SANTA MARGHERITA USA INC	100749	SANTA MARGHERITA P/GRIG ALTO - 375ML	WINE	0.08	1.0	1.0	January	2020-01-01	1.08
7	2020	1	BROWN-FORMAN BEVERAGES WORLDWIDE	1008	JACK DANIELS COUNTRY COCKTAIL SOUTHERN	BEER	0.00	0.0	2.0	January	2020-01-01	2.00

```
df['TOTAL SALES'].sum()
```

```
np.float64(1028499.3899999999)
```

```
df['RETAIL SALES'].sum()
```

```
np.float64(204223.46000000002)
```

```
df['WAREHOUSE SALES'].sum()
```

```
np.float64(824275.9299999999)
```

```
monthly_sales = df.groupby('MONTH_NAME')['TOTAL SALES'].sum()
monthly_sales
```

MONTH_NAME	TOTAL SALES
January	358963.32
July	510087.02
March	147227.46
September	12221.59

dtype: float64

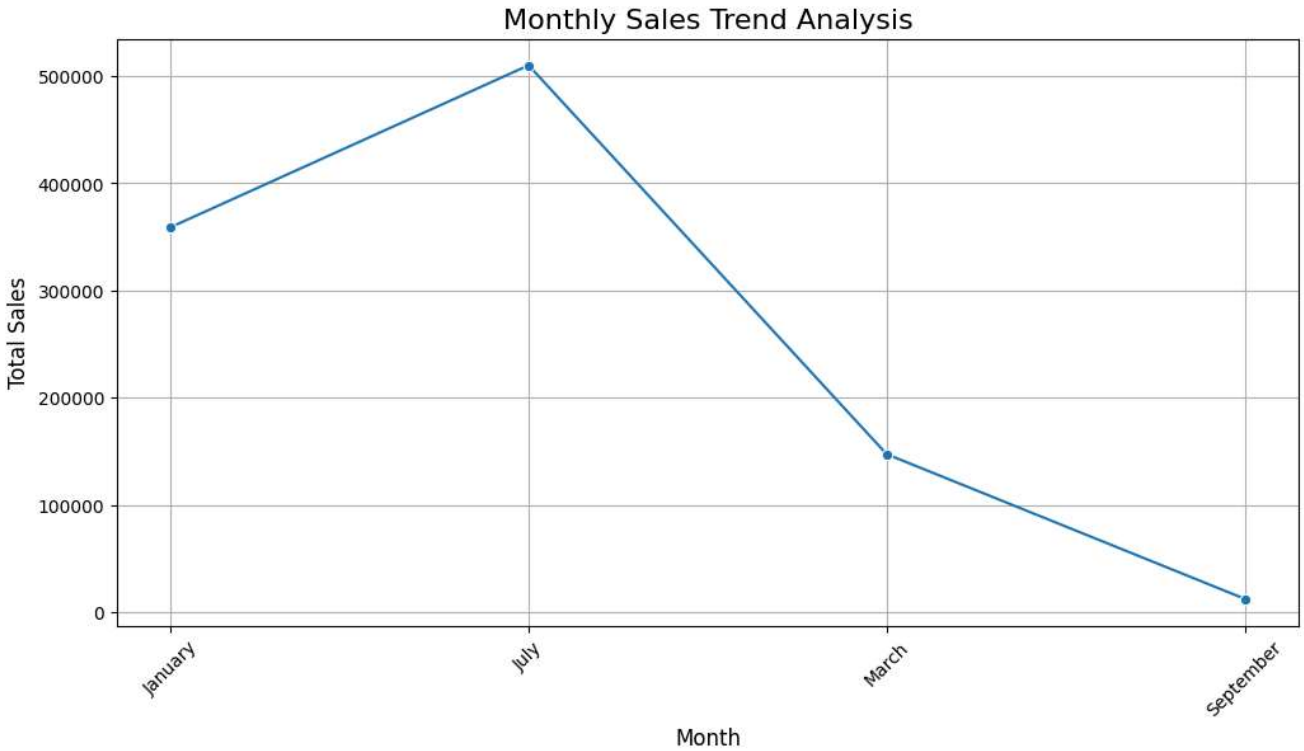
Start coding or generate with AI.

```
monthly_sales = df.groupby('MONTH_NAME')['TOTAL SALES'].sum().reset_index()

plt.figure(figsize=(12,6))
sns.lineplot(data=monthly_sales, x='MONTH_NAME', y='TOTAL SALES', marker='o')

plt.title("Monthly Sales Trend Analysis", fontsize=16)
plt.xlabel("Month", fontsize=12)
plt.ylabel("Total Sales", fontsize=12)
plt.xticks(rotation=45)

plt.grid(True)
plt.show()
```

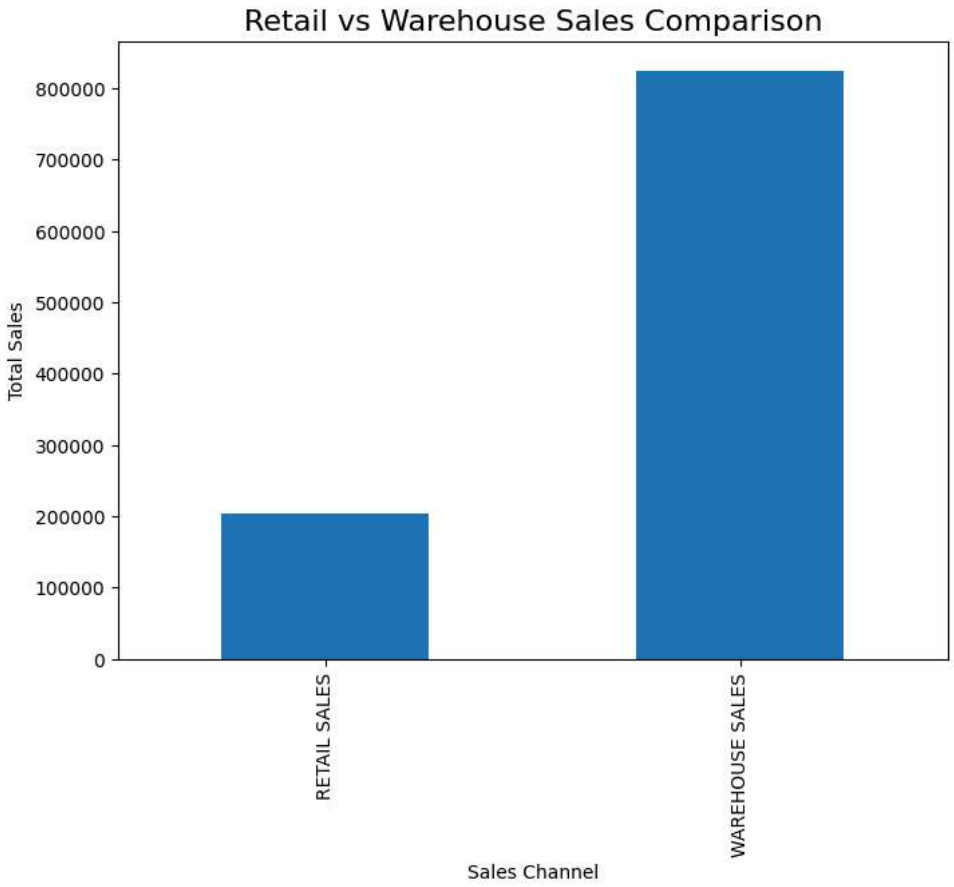



```
sales_channel = df[['RETAIL SALES','WAREHOUSE SALES']].sum()
```

```
plt.figure(figsize=(8,6))
sales_channel.plot(kind='bar')

plt.title("Retail vs Warehouse Sales Comparison", fontsize=16)
plt.xlabel("Sales Channel")
plt.ylabel("Total Sales")

plt.show()
```

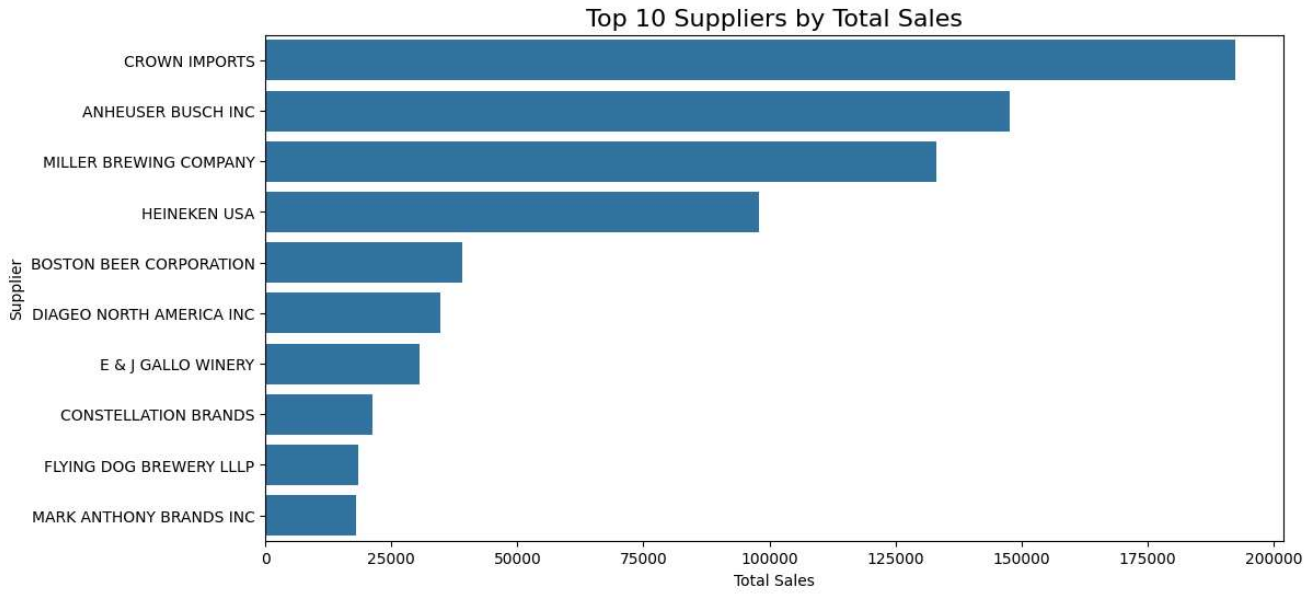



```
top_suppliers = df.groupby('SUPPLIER')['TOTAL SALES'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(12,6))
sns.barplot(x=top_suppliers.values, y=top_suppliers.index)

plt.title("Top 10 Suppliers by Total Sales", fontsize=16)
plt.xlabel("Total Sales")
plt.ylabel("Supplier")

plt.show()
```

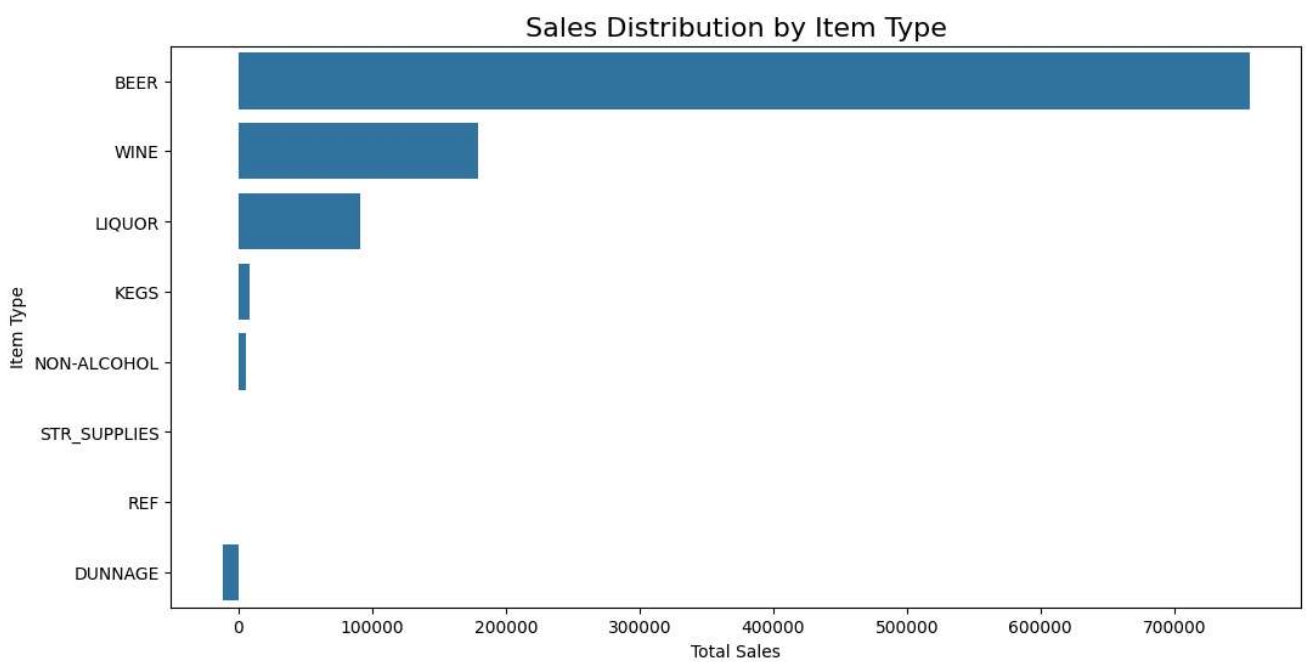



```
item_sales = df.groupby('ITEM TYPE')['TOTAL SALES'].sum().sort_values(ascending=False)

plt.figure(figsize=(12,6))
sns.barplot(x=item_sales.values, y=item_sales.index)

plt.title("Sales Distribution by Item Type", fontsize=16)
plt.xlabel("Total Sales")
plt.ylabel("Item Type")

plt.show()
```



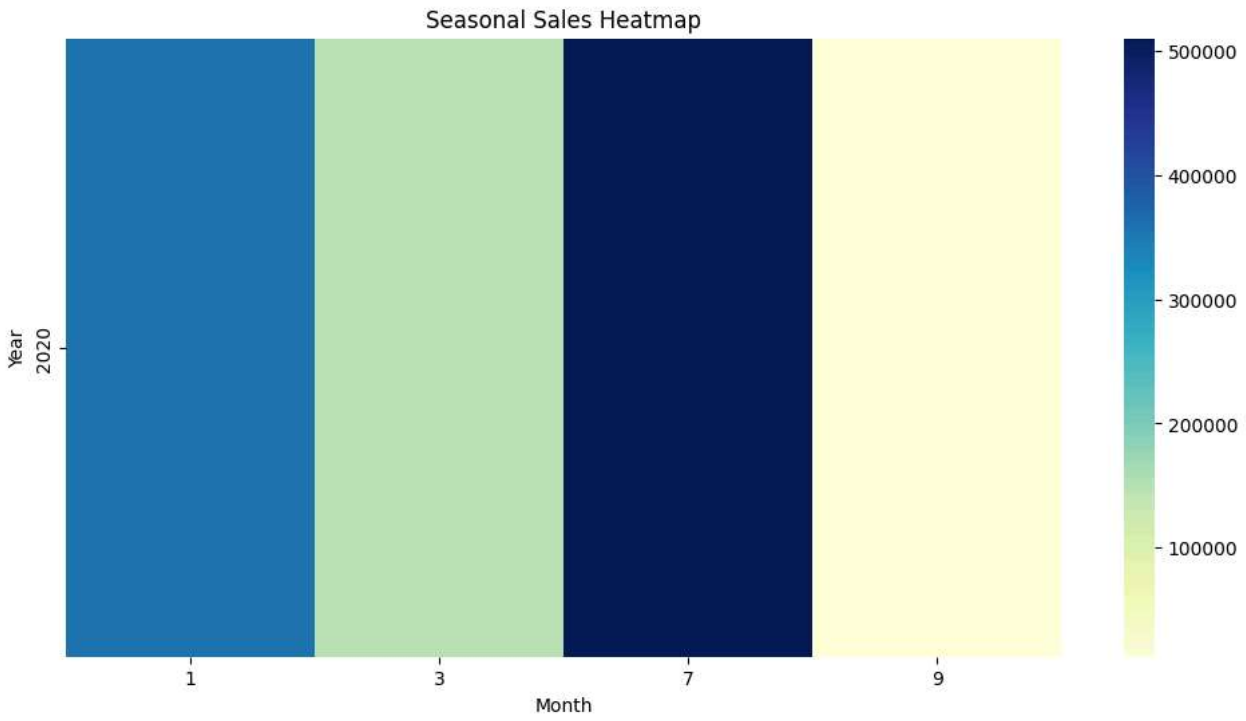

```
pivot_table = df.pivot_table(values='TOTAL SALES', index='YEAR', columns='MONTH', aggfunc='sum')

plt.figure(figsize=(12,6))
sns.heatmap(pivot_table, cmap="YlGnBu", annot=False)

plt.title("Seasonal Sales Heatmap")
plt.xlabel("Month")
```

```
plt.ylabel("Year")

plt.show()
```



Start coding or generate with AI.

```
df.to_csv('Retail_Cleaned_Data.csv', index=False)
```

```
from google.colab import files
files.download('Retail_Cleaned_Data.csv')
```

Start coding or generate with AI.